

RESEARCH STATEMENT - LUCAS GAGNON

Polynomials provide a common language across geometry, combinatorics, and representation theory. Distinguished families, including symmetric, Schubert, and Macdonald polynomials, form a “core vocabulary” that communicates information between these fields. As an algebraic combinatorialist, I develop explicit mechanisms such as summation formulas, multiplication rules, and coefficient interpretations that enhance our understanding of these families.

My research is organized around the question of geometric interpretation: to what extent is the combinatorial behavior of a polynomial family encoded in the cohomology of an associated algebraic variety? My central contributions introduce Catalan objects (e.g. non-crossing partitions and parking functions) as a combinatorial bond between polynomials and geometry. Two principal programs develop this perspective:

- (1) *Quasisymmetry and Schubert calculus.* Quasisymmetric functions occur throughout enumerative combinatorics and representation theory. In joint work [6, 7, 8, 10], my collaborators and I define a geometric analogue of Schubert calculus for quasisymmetric polynomials, providing a canonical geometric realization sought in [1, 5, 40, 44, 39]. Our result connects quasisymmetry and Schubert calculus to Coxeter–Catalan combinatorics, which we further explore in [9]. In ongoing work [11] and [23], we will follow these connections further to address questions about interpreting related polynomial families [48], the geometry of Richardson varieties [33, 35] and degenerations of Hessenberg varieties [2, 4, 31].
- (2) *Macdonald theory via affine Springer fibers.* Macdonald polynomials link representation theory with harmonic analysis, algebraic geometry, and mathematical physics. Building on explicit formulas for nonsymmetric Koornwinder polynomials in the preprint [16] and a study of Cherednik-algebra Springer representations in the in-progress manuscript [18], this program aims to create computable formulas across Lie types using affine Springer geometry. These constructions take up the question raised in [46] about geometric interpretation of Macdonald polynomials, and open a route for extending type- A (q, t) -Catalan and shuffle phenomena [36, 24] across Lie types.

Together with my earlier work on unipotent algebraic groups [20, 21] and their connection to chromatic symmetric functions [22], these programs form a broader study of Catalan combinatorics in flag geometry. This also informs my practice of undergraduate research mentorship, which has resulted in the published article [3] and in-preparation manuscript [13].

1. FLAG GEOMETRY AS A POLYNOMIAL DICTIONARY

Both research programs are informed by the geometry of the *flag variety*

$$\mathrm{Fl}_n = \{0 \subsetneq F_1 \subsetneq \cdots \subsetneq F_n = \mathbb{C}^n\} \cong \mathrm{GL}_n/B \quad \text{for} \quad B = \{\text{upper triangular } g \in \mathrm{GL}_n\},$$

which is a smooth, projective algebraic variety. The permutation matrices $w \in S_n \subseteq \mathrm{GL}_n$ correspond to coordinate subspace flags $wB/B \in G/B$, and the closure of the B -orbit

BwB/B is called the *Schubert variety* $X_w \subseteq \mathrm{Fl}_n$. The classes $[X_w]$ form a homology basis, so the Kronecker dual cohomology classes give a basis for $H^\bullet(\mathrm{Fl}_n; \mathbb{Z})$.

Borel's presentation identifies $H^\bullet(\mathrm{Fl}_n; \mathbb{Z})$ with the coinvariant ring

$$H^\bullet(\mathrm{Fl}_n; \mathbb{Z}) \cong \mathbb{Z}[\mathbf{x}_n] / \langle \mathbb{Z}[\mathbf{x}_n]_+^{S_n} \rangle, \quad \text{for} \quad \mathbf{x}_n = \{x_1, \dots, x_n\}, \quad (1)$$

where $\mathbb{Z}[\mathbf{x}_n]_+^{S_n}$ denotes the symmetric polynomials with no constant term. Each Schubert class has a combinatorially-defined representative in the presentation, the Schubert polynomial $\mathfrak{S}_w \in \mathbb{Z}[\mathbf{x}_n]$. For any cohomology class ψ , one can ask for the *Schubert expansion*

$$\psi = \sum_w a_w^\psi \mathfrak{S}_w. \quad (2)$$

If ψ is Poincaré dual to a subscheme $Y \subseteq \mathrm{Fl}_n$ then the a_w^ψ will be nonnegative, and computing their value combinatorially is an interesting and nontrivial problem.

A complementary approach known as a Goresky–Kottwitz–MacPherson (GKM) theory [25] comes from the action of the torus T of diagonal matrices. The moment graph of 0- and 1-dimensional T -orbits in Fl_n is isomorphic to the Cayley graph $\mathrm{Cay}(S_n)$ for the transposition generators (ij) . Then [25] presents the T -equivariant cohomology $H_T^\bullet(\mathrm{Fl}_n) \cong H_T^\bullet(\mathrm{Cay}(S_n))$, where

$$H_T^\bullet(\mathrm{Cay}(S_n)) = \left\{ f : S_n \rightarrow \mathbb{Z}[\mathbf{t}_n] \mid t_i - t_j \text{ divides } f_w - f_v \text{ if } w \xrightarrow{(ij)^-} v \right\}. \quad (3)$$

This is a free module over $H_T^\bullet(pt) \cong \mathbb{Z}[\mathbf{t}_n]$ with special fiber $H^\bullet(\mathrm{Fl}_n; \mathbb{Z})$ and generic fiber equal to an S_n -orbit of points in $\mathbb{C}^n$, a phenomenon also known as “orbit harmonics.” After adjoining inverses $(t_i - t_j)^{-1}$ for $i \neq j$, the point-classes $\delta_v = (\delta_{v,w})_{w \in S_n}$ become a basis, and each equivariant class ψ therefore has a rational function-coefficient expansion

$$\psi = \sum_{w \in S_n} m_w^\psi(\mathbf{t}_n) \delta_w. \quad (4)$$

If ψ is the equivariant Poincaré dual to a T -stable subscheme Y , then the coefficients m_w^ψ are related to Brion's equivariant multiplicities [14].

2. QUASISYMMETRY AND SCHUBERT CALCULUS

A polynomial $f \in \mathbb{Z}[x_1, \dots, x_n]$ is *quasisymmetric* if the coefficient of a monomial $x_{i_1}^{\alpha_1} \cdots x_{i_k}^{\alpha_k}$ with $i_1 < \cdots < i_k$ depends only on the composition $\alpha = (\alpha_1, \dots, \alpha_k)$, not on the indices $i_1, \dots, i_k$. Quasisymmetric polynomials were first introduced as an enumerative tool via the theory of P -partitions, but now play many other roles in algebraic combinatorics and combinatorial representation theory. Inspired by Borel's presentation, Aval, Bergeron, and Bergeron [1] defined a quasisymmetric analogue of the coinvariant ring,

$$\mathrm{QSCoinv}_n = \mathbb{Z}[\mathbf{x}_n] / \langle \mathrm{QSym}_n^+ \rangle,$$

where QSym_n^+ denotes the quasisymmetric polynomials in $\mathbf{x}_n$ with no constant term. The landmark result of [1] was that the rank of $\mathrm{QSCoinv}_n$ is the n th Catalan number, $\mathrm{Cat}_n =$

$\frac{1}{n+1} \binom{2n}{n}$. This and other developments led to a search for geometric interpretations of quasisymmetric, leading to the topological and geometric models for quasisymmetric power series in [5, 40, 44].

Nadeau and Tewari [37] unexpectedly encountered QSCoinv_n while trying to find the Schubert expansion for the class of the *permutahedral variety*, a generic T -orbit closure in Fl_n . Their solution introduces a novel basis of QSCoinv_n indexed by Cat_n -many plane binary forests, henceforth known as *forest polynomials*. The paper [37] and Nadeau–Spink–Tewari [38, 39] found that forest polynomials have many Schubert-like properties including positive structure constants a divided-difference operator calculus, and individual dualities between forest polynomials and Richardson varieties, but no single geometric object that precisely represents QSCoinv_n .

The geometric solution. In [6] Bergeron and I created an orbit harmonics phenomenon for QSCoinv_n using the noncrossing partitions of the cycle $c = (n \cdots 21)$, a canonically defined set $\text{NC}_c \subseteq S_n$ with Cat_n -many elements [34]. This combinatorial insight led to a complete solution: in two papers with Bergeron, Nadeau, Spink, and Tewari [7, 8], I define the *quasisymmetric flag variety*

$$\text{QFl}_n = \bigcup_{w \in S_n} w^{-1} X_w^{wc} \subseteq \text{Fl}_n,$$

where $w^{-1} X_w^{wc}$ denotes the translation of a Richardson variety X_w^{wc} by $w^{-1} \in \text{GL}_n$. While unusual, the translation is a key part of our construction, principally because it makes the torus-fixed point set of QFl_n precisely NC_c .

Theorem 1 ([8, Theorem A]). *We have $H^\bullet(\text{QFl}_n; \mathbb{Z}) \cong \text{QSCoinv}_n$ and using the notation of Equation (3), $H_T^\bullet(\text{QFl}_n; \mathbb{Z}) \cong H_T^\bullet(\text{Cay}(\text{NC}_c))$.*

Previously [39] established a duality between forest polynomials and the untranslated X_w^{wc} ; Theorem 1 transforms this into an interpretation of forest polynomials as the “Schubert polynomials” of QFl_n .

Extensions. In a submitted preprint with Bergeron, Spink, and Tewari [9], I found a Lie type generalization of the noncrossing partition combinatorics of [8]. For a reductive group G , a *Coxeter element* c is any product of the simple generators of the Weyl group W . Each Coxeter element c determines a set of c -noncrossing partitions $\text{NC}_c \subseteq W$ counted by Cat_W , the W -Catalan number. For every G and c , the paper [9] constructs a *Coxeter flag variety* $\text{CFl}_c \subseteq G/B$ as a complex of Richardson varieties that admits Plücker vanishing conditions and GKM equivariant cohomology presentations via NC_c . While $H^\bullet(\text{CFl}_c)$ does not in general admit a coinvariant presentation with Schubert polynomials, we find a Coxeter–Schubert basis in the GKM ring with positive multiplication.

In another preprint [10], Bergeron, Spink, Tewari and I consider the image of QFl_n under the projection from Fl_n to the Grassmannian $\text{Gr}(r, n)$. We call the resulting space the *quasisymmetric Grassmannian* $\text{QGr}(r, n)$ and show that it has cohomology realizing a finite truncation of QSym_r and a Schubert-type basis given by fundamental quasisymmetric polynomials. This gives a distinct interpretation of quasisymmetry from [5, 40, 44].

New directions. My collaboration with Bergeron, Spink, and Tewari is ongoing and making progress in two directions. In a pair of papers (in preparation) we will apply our

paradigm for geometric interpretation to Stembridge’s peak algebra [48]. The first work in this series is the in-preparation manuscript [11], which will construct “peak forest polynomials” with positive product rules, Schubert class expansion, and divided-difference calculus. A sequel [12] will construct a “peak quasisymmetric flag variety” as a toric complex in the isotropic flag variety; this offers a distinct type B/C generalization of our CFl_c construction. Later work will dig deeper into CFl_c for $G = \text{GL}_n$ and any $c \neq (n \cdots 21)$; here we obtain new geometric bases of QSCoinv_n indexed by Cambrian forests [45].

Another direction is Hessenberg generalization: both Nadeau–Tewari’s initial work and our later joint work rely on properties of the permutahedral variety that also hold for the wider family of Hessenberg varieties. A natural question is whether previous results are special cases of a more general Hessenberg phenomenon. My in-preparation work with Woo and Zhuang [23] is a proof of concept, using forest polynomials to compute the Schubert expansion of all irreducible components of every *subregular* Hessenberg variety with the same discrete (root ideal) parameter. Future work will explore other aspects of this question.

3. MACDONALD THEORY VIA AFFINE SPRINGER GEOMETRY

Each affine root system determines a family of *Macdonald polynomials*, which are symmetric Laurent polynomials P_μ with rational coefficients in two to six additional parameters. Macdonald theory studies the P_μ alongside objects like nonsymmetric Macdonald polynomials E_μ , which de-symmetrize the P_μ , and specializations including the Jack (Heckman–Opdam) [17], Hall–Littlewood [46], and Demazure [32] polynomials.

There is an elegant characterization of the P_μ , E_μ and specializations as joint eigenfunctions of Cherednik–Dunkl operators in the polynomial representation of the double affine Hecke algebra (DAHA) $\mathbf{H}$. Vasserot further shows that (quotients of) this $\mathbf{H}$ -representation can be realized on the K -groups of affine Springer fibers [49]. However, neither result is particularly constructive, and progress on explicit formulas for the E_μ and P_μ [46] and their geometric meaning [42, 49] have largely developed on separate tracks.

Type A serves as a template for satisfactory answers. Here there is a robust literature of combinatorial formulas [15, 28, 29], a bespoke (co)homological theory of Hilbert schemes [27] and knot homology [41], and work connecting geometry and combinatorics [30, 36].

A combinatorial benchmark. Koornwinder polynomials are Macdonald polynomials for the nonreduced affine root system $(C_n^\vee, C_n)$, and the Macdonald polynomials for each classical root system can be expressed as limits and/or special cases of Koornwinder polynomials. Key work on Koornwinder polynomials has been limited to algebraic results via the action of $\mathbf{H}$ [47], type-independent alcove-walk formulas with extremely fast term growth [43, 46], with smaller tableau formula known only in special cases [19].

My joint work with Colmenarejo and Ram [16] establishes a new formula for arbitrary nonsymmetric Koornwinder polynomials

$$E_\mu = \text{nf}(\mu) \sum_{\substack{\text{compressed set-valued} \\ \text{tableaux } T \text{ of shape } \mu}} \text{wt}(T) x^{\text{end}(T)}.$$

for weights wt , end and normalization nf . This formula has significantly fewer terms than any previous formula and introduces a uniform combinatorics of “Koornwinder tableaux.”

While this work is purely combinatorial, it will provide a combinatorial anchor for future geometric developments in affine flag varieties of (possibly twisted) classical type.

Affine Springer Theory. The next step is to use explicit geometric constructions to gain insight about Macdonald polynomials in general type. Let $F = \mathbb{C}((\epsilon))$. Every loop group $\tilde{G} = G(F)$ has a Borel-like *Iwahori subgroup* I , and the affine flag variety is the ind-scheme $\tilde{G}/I$; for $\tilde{G} = \mathrm{GL}_n(F)$, this space parametrizes n -periodic flags of ϵ -lattices in F^n . Each $\gamma \in \mathrm{Lie}(\tilde{G})$ determines an *affine Springer fiber*

$$\mathrm{Sp}_\gamma = \{gI \in \tilde{G}/I \mid g^{-1}\gamma g \in \mathrm{Lie}(I)\}.$$

For γ an elliptic homogeneous element, Sp_γ admit the action of a special torus T_γ , and Varagnolo–Vasserot use equivariant methods to construct an $\mathbf{H}$ -representations on the K -groups $K_\bullet(\mathrm{Sp}_\gamma^{T_\gamma})$ [49].

The nonsymmetric Heckman–Opdam polynomials are the limits

$$E_\lambda^{(c)}(X; \nu) = \lim_{r \rightarrow 0} E_\lambda(X; q^{\delta r}, t_\alpha^{c_\alpha u r}),$$

which form eigenbases for modules of the degenerate DAHA $\mathbf{H}'$ (the ‘trigonometric Cherednik algebra’). In type A these are nonsymmetric Jack polynomials. As a cohomological analogue of the DAHA, [49] show that $\mathbf{H}'$ acts on the Borel–More homology $H_\bullet^{BM}(\mathrm{Sp}_\gamma^{T_\gamma})$, while more recently [42] have proved a similar result for the localization of the equivariant cohomology $H_{T_\gamma}^\bullet(\mathrm{Sp}_\gamma)$.

In the in-preparation work [18], Ram, Colmenarejo, Thiem, and I will give an intrinsic description of the Macdonald polynomial K -classes by first proving a similar degenerate result. We will focus on a particularly nice family known as the *Coxeter* elliptic homogeneous affine Springer fibers, which carry robust combinatorics in terms of the dilation dA_0 of the fundamental alcove A_0 . These Springer fibers admit a Schubert-style affine paving with cells, and T_γ -fixed points have the for wI over the affine Weyl group elements w for which $wA_0 \subseteq dA_0$ [42]. In type A these have a natural bijection to parking functions [26]. This makes explicit descriptions of their cohomology a nontrivial, but computable problem via equivariant localization $H_{T_\gamma}^\bullet(\mathrm{Sp}_\gamma) \hookrightarrow H_{T_\gamma}^\bullet(\mathrm{Sp}_\gamma^{T_\gamma}) \cong \prod_{wA_0 \in dA} H_{T_\gamma}^\bullet(pt)$, though we do not get a full GKM-style presentation.

Expected Theorem 2 ([18], in progress). *For γ elliptic homogeneous of Coxeter type, $H_{T_\gamma}^\bullet(\mathrm{Sp}_\gamma^{T_\gamma})$ affords an antispherical representation of $\mathbf{H}'$, and the Jack eigenclasses are $\{m_w \delta_w \in H_{T_\gamma}^\bullet(\mathrm{Sp}_\gamma^{T_\gamma}) \mid wA_0 \in dA_0\}$ for explicit coefficients m_w .*

Note that the representation itself is known and not the novel part of Expected Theorem 2, which is instead the interpretation of Jack classes and the forthcoming combinatorial description of the coefficients m_w . To extend this result to a Macdonald polynomial interpretation, the next step is the highly nontrivial task of lifting, via Borel–Moore homology, a cohomological basis statement to K -theory. Fortunately, the representation-theoretic degeneration from the DAHA to the degenerate DAHA carries more structure than the ordinary Chern character map, resolving genuine ambiguities into challenging problems with unique solutions.

Further work will deepen the computational reach of this interpretation, for example giving a bijective rule for the distribution of cohomological and perverse gradings found in [42], which should lead to analogues of (q, t) -Catalan numbers. Moreover, Expected Theorem 2 leads to the surprising realization that transitioning from Jack (or Macdonald) classes to Schubert classes is inherently geometric, through equivariant multiplicity, and it will be interesting to explore the meaning of this on a polynomial level.

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